

Demo Walkthrough — the live system, on camera

SaResQ · SIH26177 · target 4:00 · read the pre-flight page BEFORE you record

This is not the slide script. `SaResQ_Pitch_Script.pdf` narrates the six slides; this drives the running payload and the ground station in front of a lens. Two things make or break the take and neither is about what you say: **the room temperature** and **the network**. Both are handled on this page.

Pre-flight — do all of this before you hit record

The room must be cooler than you are.

Measured on 24 Sep: ambient 32 °C, blobs 33.5–34.3 °C. That is the crossover band — skin and background almost equal — and the gate had almost nothing to work with (z scattered 2.7–4.2, blobs drifting to the frame edges).

Film at night, or in an air-conditioned room, or early morning.

Aim for ambient below 28 °C: it takes peak z from ~4 to comfortably past 6, the blob sits solidly on the body instead of wandering, and the thermal panel actually looks like a person. This single choice does more for the footage than anything else in this document.

1 • Bring the payload up and find it

```
ping -c2 saresq.local          # often fails even when the Pi is fine
for i in $(seq 1 254); do echo 192.168.1.$i; done | \
xargs -P128 -I{} sh -c 'nc -z -w1 {} 8091 2>/dev/null && echo FOUND {}'
```

mDNS has failed mid-session twice while the Pi was healthy and serving. Never let the demo depend on `saresq.local` — scan for it, then use the raw IP.

2 • Start the ground station against that IP

```
cd ~/saresq
.venv/bin/python -m saresq.dashboard.app --payload <PI-IP> --port 5050
```

3 • Verify before you waste a take

```
curl -s http://127.0.0.1:5050/api/live | python3 -m json.tool | head -20
```

- ☐ `connected: true` and `age_s` under 2
- ☐ `rgb_blind: false` — if true, put a lamp on. A dark crop makes the banner hedge and it reads badly on camera
- ☐ `fired: true` when you step in front of the thermal camera
- ☐ Thermal `spread` above ~5 °C on `:8091/stats` — below that the scene is too uniform to be interesting
- ☐ Evidence page empty — power-cycle the payload for a clean mission so the demo starts from zero rather than yesterday's crops
- ☐ Close other apps. The Pi runs at 60–68 °C under load and the laptop is running a second detector

4 • Set a position, because there is no GPS fix indoors

The map is honest about having no satellite fix, which is correct but looks empty on camera. On the Map page, **shift-click** where you actually are, or press **Use this device**. It is tagged as an operator datum and drawn differently from a measured fix — say that out loud when you show it. It is a strength, not an apology.

The take — eight shots, about four minutes

01 The thing itself

0:00 – 0:25

SHOW The payload in your hand. Turn it over — Pi, thermal array, camera, GNSS mast, the printed tray.

SAY *"This is SaResQ. Everything that finds a survivor happens on this board — a Raspberry Pi 4, a thermal array, a camera. About thirteen thousand rupees, three hundred grams, and it bolts onto any F450-class drone. No cloud, no mobile network. The disasters we built it for take the network down first."*

02 Thermal first — the whole idea in one shot

0:25 – 1:05

SHOW /feed . Both panels side by side. Walk into frame. Gate fires, banner changes, blob lights up on the 32×24.

SAY *"Most drone detection runs a neural network across the whole picture and hopes the person is big enough and lit well enough to find. At night, in smoke, in an unlit room, that fails — and that is exactly where rescue happens. So we invert it. A person is a thermal object first. This thirty-two by twenty-four array — seven hundred and sixty-eight pixels, no AI at all — asks one question of every frame: what here is anomalously warm compared to the rest of this scene, right now?"*

Let the thermal panel breathe for a beat. The orange body shape is the most legible image in the whole video.

03 The cascade — why it runs on a Pi at all

1:05 – 1:45

SHOW Point at the orange crop boxes on the detector panel, then the green detection box and its score.

SAY *"Only those regions get handed to the neural network. Two to six small crops, not the whole frame. We measured both on this hardware: full-frame costs five hundred and thirty-two milliseconds, a crop costs twenty-eight. That nineteen-fold difference is the entire reason a real detector runs in real time on a four-thousand-rupee computer with no GPU."*

04 The two-sided gate — the shot nobody else will have

1:45 – 2:25

SHOW **Set this up in advance:** a hot surface at 45–55 °C in frame — a pan of hot water, an induction plate, a laptop charger brick. Hold your hand flat in front of it. On the thermal panel your hand is now *darker* than the background, and the gate still fires.

SAY *"Here is the part built for India specifically. Skin sits around thirty-one to thirty-four degrees. Sun-baked concrete and asphalt in an Indian summer reach fifty to sixty — hotter than a person. So a survivor lying on that ground is not a warm spot at all. They are a cold spot. Every detector built on 'people are warmer than their background' goes blind exactly when heatstroke makes finding them most urgent. Our test fires on deviation in either direction — you can see the gate holding on my hand even though my hand is the cooler thing in this frame."*

If you can only shoot one clever thing, shoot this. It is a real physics argument, visible on screen, and it backs the "Designed for India" claim with something other than a slogan.

05 What the system will and will not claim

2:25 – 3:00

SHOW Three states, on camera, in order. **(a)** Stand still — banner holds body heat. **(b)** Move — it goes to a live person. **(c)** Cover the RGB lens with your hand while staying in thermal view — it says the camera is blind rather than reporting nobody there.

SAY *"Motion is what separates a survivor from a warm object — a car bonnet is warm, a sun-heated slab is warm. Something that changes shape between frames is breathing. But notice what it does not say when I stand still: it does not call it a weak result. An unconscious casualty is the most urgent person on any site and they do not move. And when I cover the camera, it tells the operator the camera is blind — not 'no person found'. Those two sentences mean opposite things to a rescue team."*

06 Hazards and the map

3:00 – 3:30

SHOW `/hazards` — the live class bars. Then `/map` with your datum set and a contact on it.

SAY *"Alongside the survivor search, a second classifier scores the scene for fire, flooding and structural collapse — so a team knows not just where someone is, but what they would be walking into. And every find is geo-tagged. There is no satellite fix indoors, so this position is an operator datum — and the map draws it differently from a measured fix. It never dresses an assumption up as a measurement."*

07 Evidence and the second opinion

3:30 – 3:55

SHOW `/evidence` — crops captured during this session. Then `/review` — the triage queue with confidence and coordinates.

SAY *"Every find is kept as evidence with its crop, its thermal patch and its coordinates, ordered so the most urgent reaches an operator first. The ground station also runs a larger model over the same imagery as an independent second opinion — and it is allowed to promote something the drone nearly discarded, but never to bury it. Missing a survivor and raising a false alarm are not the same size of mistake."*

08 Close

3:55 – 4:10

SAY *"Everything you just saw ran with no internet, on hardware costing about thirteen thousand rupees, against enterprise thermal drones at three to twenty-two lakh. Perception is solved and measured. GPS-denied navigation is what we are building onto it next — on a proven open-source flight stack, from a base that already works."*

Do not show these

THING	WHY	INSTEAD
Frame-rate numbers	Thermal accepts ~1.8 fps because 61% of sensor reads are discarded as torn. Correct behaviour, bad optics unprompted.	Don't raise it. If asked later, it is a sensor read-synchronisation limit, not a compute one.
Analytics page	Carries model-comparison detail that invites questions you do not need in a four-minute video.	Skip. Evidence and Review tell the story better.
The Pi in thermal view	It runs at 60–68 °C and reads as a hot blob competing with you.	Keep the board out of the thermal camera's field.
An empty Evidence page	Reads as broken.	Trigger two or three captures before recording that section.

If it breaks mid-take

- **Banner says link down** — almost always mDNS, not the Pi. Check `:8091/stats` in a browser; if that answers, restart the ground station against the raw IP.
- **Counts look wrong** — `n_air` and `n_ground` in `/api/live` show which machine said what.
- **Gate will not fire** — the room is too warm. See the pre-flight box. No software setting fixes a scene with no thermal contrast in it.
- **Payload unresponsive** — `sudo systemctl restart saresq-demo.service`, give it eight seconds.
- **Shutting down** — always `sudo systemctl stop saresq-demo.service && sudo sync && sudo shutdown -h now`. Pulling power risks the SD card.

Numbers you can quote, all measured on this hardware

CLAIM	VALUE
Full-frame detector on the Pi	531.9 ms
One cascade crop	28.2 ms
Designed cascade against frame budget	177.5 / 250 ms

Thermal sensor raw rate	4.67 fps
Gate threshold	$ z \geq 2.5$, two-sided
Thermal–RGB calibration residual	0.506 thermal px
False motion removed by the deadband	23 in 80 still frames
Smallest real motion detected	4 px at 1.55 K
Ground model false-positive rate	0.013
Payload mass / cost	~300 g / ~₹13,000
Automated tests	230 passing

Pair with [SaResQ_Pitch_Script.pdf](#) (slide narration) and [SaResQ_QA_Prep.pdf](#) (judge questions). Every figure above was measured on the payload, not taken from a datasheet — if a number is not in this table, do not say it on camera.